17. (a) Let $0 < p_1 < p_2 < 1$. Define $A_i = \{x : F(x) \geq p_i\}$ for $i = 1, 2$. Since $p_1 < p_2$ and F is nondecreasing, it follows that $A_2 \subset A_1$. Hence, the smallest number in A_1 (which equals $F^{-1}(p_1)$ by definition) is no greater than the smallest number in A_2 (which equals $F^{-1}(p_2)$ by definition. That is, $F^{-1}(p_1) \leq F^{-1}(p_2)$, and the quantile function is nondecreasing.

    (b) Let $x_0 = \lim_{\substack{p \to 0 \\ p>0}} F^{-1}(p)$. We are asked to prove that x_0 is the greatest lower bound of the set $C = \{c : F(c) > 0\}$. First, we show that no $x > x_0$ is a lower bound on C. Let $x > x_0$ and $x_1 = (x + x_0)/2$. Then $x_0 < x_1 < x$. Because $F^{-1}(p)$ is nondecreasing, it follows that there exists $p > 0$ such that $F^{-1}(p) < x_1$, which in turn implies that $p \leq F(x_1)$, and $F(x_1) > 0$. Hence $x_1 \in C$, and x is not a lower bound on C. Next, we prove that x_0 is a lower bound on C. Let $x \in C$. We need only prove that $x_0 \leq x$. Because $F^{-1}(p)$ is nondecreasing, we must have $\lim_{\substack{p \to 0 \\ p>0}} F^{-1}(p) \leq F^{-1}(q)$ for all $q > 0$. Hence, $x_0 \leq F^{-1}(p)$ for all $q > 0$. Because $x \in C$, we have $F(x) > 0$. Let $q = F(x)$ so that $q > 0$. Then $x_0 \leq F^{-1}(q) \leq x$. The proof that x_1 is the least upper bound on the set of all d such that $F(d) < 1$ is very similar.

    (c) Let $0 < p < 1$. Because F^{-1} is nondecreasing, $F^{-1}(p^-)$ is the least upper bound on the set $C = \{F^{-1}(q) : q < p\}$. We need to show that $F^{-1}(p)$ is also that least upper bound. Clearly, $F^{-1}(p)$ is an upper bound, because F^{-1} is nondecreasing and $p > q$ for all $q < p$. To see that $F^{-1}(p)$ is the least upper bound, let y be an upper bound. We need to show $F^{-1}(p) \leq y$. By definition, $F^{-1}(p)$ is the greatest lower bound on the set $D = \{x : F(x) \geq p\}$. Because y is an upper bound on C, it follows that $F^{-1}(q) \leq y$ for all $q < p$. Hence, $F(y) \geq q$ for all $q < p$. Because F is nondecreasing, we have $F(y) \geq p$, hence $y \in D$, and $F^{-1}(p) \leq y$.

18. We know that $\Pr(X = c) = F(c) - F(c^-)$. We will prove that $p_1 = F*(c)$ and $p_0 = F(c^-)$. For each $p \in (0, 1)$ define

$$C_p = \{x : F(x) \geq p\}.$$

    Condition (i) says that, for every $p \in (p_0, p_1)$, c is the greatest lower bound on the set C_p. Hence $F(c) \geq p$ for all $p < p_1$ and $F(c) \geq p_1$. If $F(c) > p_1$, then for $p = (p_1 + F(c))/2$, $F^{-1}(p) \leq c$, and condition (iii) rules this out. So $F(c) = p_1$. The rest of the proof is broken into two cases. First, if $p_0 = 0$, then for every $\epsilon > 0$, c is the greatest lower bound on the set C_ϵ. This means that $F(x) < \epsilon$ for all $x < c$. Since this is true for all $\epsilon > 0$, $F(x) = 0$ for all $x < c$, and $F(c^-) = 0 = p_0$. For the second case, assume $p_0 > 0$. Condition (ii) says $F^{-1}(p_0) < c$. Since $F^{-1}(p_0)$ is the greatest lower bound on the set C_{p_0}, we have $F(x) < p_0$ for all $x < c$. Hence, $p_0 \geq F(c^-)$. Also, for all $p < p_0$, $p \leq F(c^-)$, hence $p_0 \leq F(c^-)$. Together, the last two inequalities imply $p_0 = F(c^-)$.

19. First, we show that $F^{-1}(F(x)) \leq x$. By definition $F^{-1}(F(x))$ is the smallest y such that $F(y) \geq F(x)$. Clearly $F(x) \geq F(x)$, hence $F^{-1}(F(x)) \leq x$. Next, we show that, if $p > F(x)$, then $F^{-1}(p) > x$. Let $p > F(x)$. By Exercise 17, we know that $F^{-1}(p) \geq x$. By definition, $F^{-1}(p)$ is the greatest lower bound on the set $C_p = \{y : F(y) \geq p\}$. All $y \in C_p$ satisfy $F(y) > (p + F(x))/2$. Since F is continuous from the right, $F(F^{-1}(p)) \geq (p + F(x))/2$. But $F(x) < (p + F(x))/2$, so $x \neq F^{-1}(p)$, hence $F^{-1}(p) > x$.

20. Figure S.3.14 has the plotted c.d.f., which equals $0.0045x^2/2$ for $0 < x < 20$. On the plot, we see that $F(10) = 0.225$.

3.4 Bivariate Distributions

Commentary

The bivariate distribution function is mentioned at the end of this section. The only part of this discussion that is used later in the text is the fact that the joint p.d.f. is the second mixed partial derivative of the

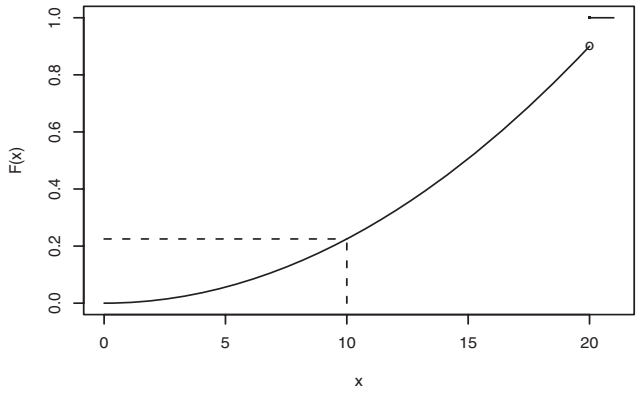


Figure S.3.14: C.d.f. for Exercise 20 of Sec. 3.3.

bivariate p.d.f. (in the discussion of functions of two or more random variables is Sec. 3.9.) If an instructor prefers not to discuss how to calculate probabilities of rectangles and is not going to cover functions of two or more random variables, there will be no loss of continuity.

Solutions to Exercises

1. (a) Let the constant value of the p.d.f. on the rectangle be c. The area of the rectangle is 2. So, the integral of the p.d.f. is $2c = 1$, hence $c = 1/2$.

   (b) $\Pr(X \geq Y)$ is the integral of the p.d.f. over that part of the rectangle where $x \geq y$. This region is shaded in Fig. S.3.15. The region is a trapezoid with area $1 \times (1 + 2)/2 = 1.5$. The integral of the

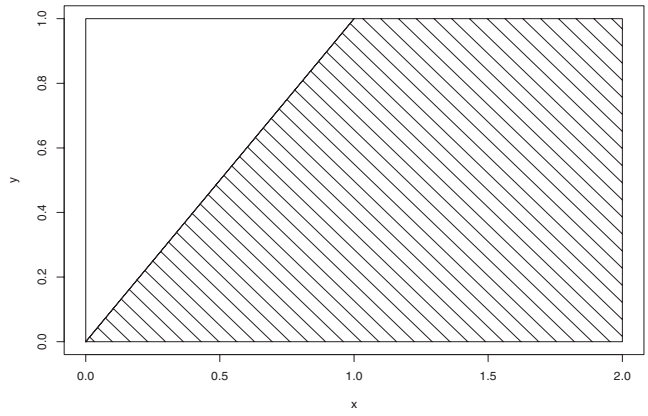


Figure S.3.15: Region where $x \geq y$ in Exercise 1b of Sec. 3.4.

   constant 1/2 over this region is then $0.75 = \Pr(X \geq Y)$.

2. The answers are found by summing the following entries in the table:

   (a) The entries in the third row of the table: 0.27.

(b) The last three columns of the table: 0.53.

(c) The nine entries in the upper left corner of the table: 0.69

(d) (0, 0), (1, 1), (2, 2), and (3, 3): 0.3.

(e) (1, 0), (2, 0), (3, 0), (2, 1), (3, 1), (3, 2): 0.25.

3. (a) If we sum $f(x, y)$ over the 25 possible pairs of values (x, y), we obtain $40c$. Since this sum must be equal to 1, it follows that $c = 1/40$.

(b) $f(0, -2) = (1/40) \cdot 2 = 1/20$.

(c) $\Pr(X = 1) = \sum_{y=-2}^{2} f(1, y) = 7/40$.

(d) The answer is found by summing $f(x, y)$ over the following pairs: $(-2, -2)$, $(-2, -1)$, $(-1, -2)$, $(-1, -1)$, $(-1, 0)$, $(0, -1)$, (0, 0), (0, 1), (1, 0), (1, 1), (1, 2), (2, 1), and (2, 2). The sum is 0.7.

4. (a) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\,dx\,dy = \int_0^1\int_0^2 cy^2\,dx\,dy = 2c/3$. Since the value of this integral must be 1, it follows that $c = 3/2$.

(b) The region over which to integrate is shaded in Fig. S.3.16.

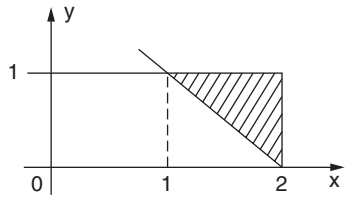


Figure S.3.16: Region of integration for Exercise 4b of Sec. 3.4.

$$\begin{aligned}\Pr(X + Y > 2) &= \iint\limits_{\substack{\text{shaded}\\ \text{region}}} f(x, y)\,dx\,dy \\ &= \int_1^2\int_{2-x}^1 \frac{3}{2}y^2\,dy\,dx = \frac{3}{8}.\end{aligned}$$

(c) The region over which to integrate is shaded in Fig. S.3.17.

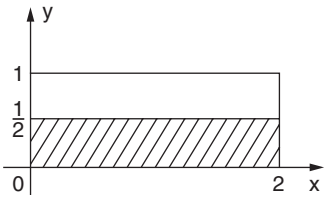


Figure S.3.17: Region of integration for Exercise 4c of Sec. 3.4.

$$\Pr\left(Y < \frac{1}{2}\right) = \int_0^2\int_0^{1/2} \frac{3}{2}y^2\,dy\,dx = \frac{1}{8}.$$

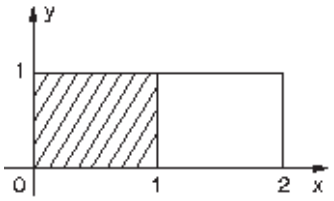


Figure S.3.18: Region of integration for Exercise 4d of Sec. 3.4.

(d) The region over which to integrate is shaded in Fig. S.3.18.

$$\Pr(X \le 1) = \int_0^1 \int_0^1 \frac{3}{2} y^2 \, dy \, dx = \frac{1}{2}.$$

(e) The probability that (X, Y) will lie on the line $x = 3y$ is 0 for every continuous joint distribution.

5. (a) By sketching the curve $y = 1 - x^2$, we find that $y \le 1 - x^2$ for all points on or below this curve. Also, $y \ge 0$ for all points on or above the x-axis. Therefore, $0 \le y \le 1 - x^2$ only for points in the shaded region in Fig. S.3.19.

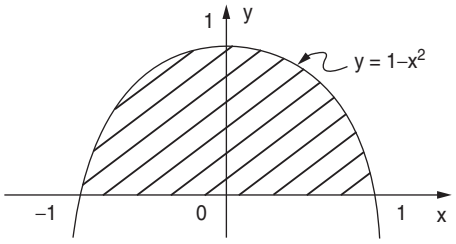


Figure S.3.19: Figure for Exercise 5a of Sec. 3.4.

Hence,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = \int_{-1}^{1} \int_{0}^{1-x^2} c\,(x^2 + y) \, dy \, dx = \frac{4}{5} c \cdot$$

Therefore, $c = 5/4$.

(b) Integration is done over the shaded region in Fig. S.3.20.

$$\Pr\left(0 \le X \le \frac{1}{2}\right) = \int\int_{\substack{\text{shaded} \\ \text{region}}} f(x, y) \, dx \, dy = \int_0^{\frac{1}{2}} \int_0^{1-x^2} \frac{5}{4}(x^2 + y) \, dy \, dx = \frac{79}{256}.$$

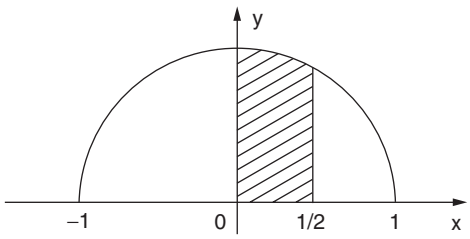


Figure S.3.20: Region of integration for Exercise 5b of Sec. 3.4.

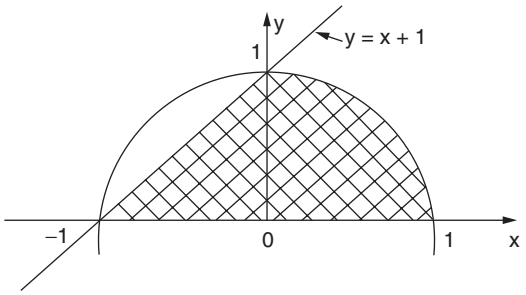


Figure S.3.21: Region of integration for Exercise 5c of Sec. 3.4.

(c) The region over which to integrate is shaded in Fig. S.3.21.

$$\begin{aligned}\Pr(Y \le X+1) &= \underset{\substack{\text{shaded}\\ \text{region}}}{\int\int} f(x,y)\,dx\,dy = 1 - \underset{\substack{\text{unshaded}\\ \text{region}}}{\int\int} f(x,y)\,dx\,dy \\ &= 1 - \int_{-1}^{0}\int_{x+1}^{1-x^2} \frac{5}{4}(x^2+y)\,dy\,dx = \frac{13}{16}.\end{aligned}$$

(d) The probability that (X, Y) will lie on the curve $y = x^2$ is 0 for every continuous joint distribution.

6. (a) The region S is the shaded region in Fig. S.3.22. Since the area of S is 2, and the joint p.d.f. is to

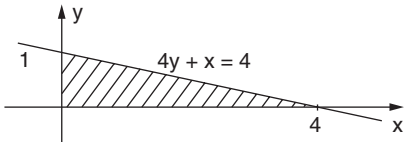


Figure S.3.22: Figure for Exercise 6a of Sec. 3.4.

be constant over S, then the value of the constant must be $1/2$.

(b) The probability that (X, Y) will belong to any subset S_0 is proportional to the area of that subset. Therefore,

$$\Pr\left[(X,Y) \in S_0\right] = \int_{S_0}\int \frac{1}{2}\,dx\,dy = \frac{1}{2}(\text{area of } S_0) = \frac{\alpha}{2}.$$

7. (a) $\Pr(X \le 1/4)$ will be equal to the sum of the probabilities of the corners (0, 0) and (0, 1) and the probability that the point is an interior point of the square and lies in the shaded region in Fig. S.3.23. The probability that the point will be an interior point of the square rather than one

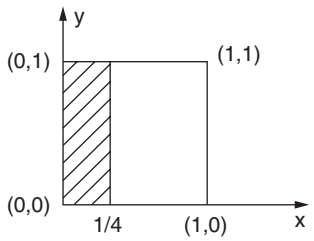


Figure S.3.23: Figure for Exercise 7a of Sec. 3.4.

of the four corners is $1-(0.1+0.2+0.4+0.1)=0.2$. The probability that it will lie in the shaded region, given that it is an interior point is 1/4. Therefore,

$$\Pr\left(X \le \frac{1}{4}\right) = 0.1 + 0.4 + (0.2)\left(\frac{1}{4}\right) = 0.55.$$

(b) The region over which to integrate is shaded in Fig. S.3.24.

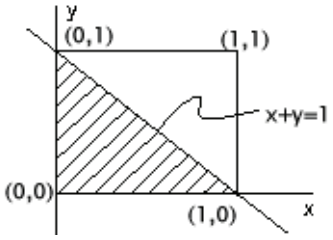


Figure S.3.24: Figure for Exercise 7b of Sec. 3.4.

$$\Pr(X+Y \le 1) = 0.1 + 0.2 + 0.4 + (0.2)\left(\frac{1}{2}\right) = 0.8.$$

8. (a) Since the joint c.d.f. is continuous and is twice-differentiable in the given rectangle, the joint distribution of X and Y is continuous. Therefore,

$$\begin{aligned} \Pr(1 \le X \le 2 \text{ and } 1 \le Y \le 2) &= \Pr(1 < X \le 2 \text{ and } 1 < Y \le 2) = \\ F(2,2) - F(1,2) - F(2,1) + F(1,1) &= \frac{24}{156} - \frac{6}{156} - \frac{10}{156} + \frac{2}{156} = \frac{5}{78} \end{aligned}$$

(b)

$$\begin{aligned} \Pr(2 \le X \le 4 \text{ and } 2 \le Y \le 4) &= \Pr(2 \le X \le 3 \text{ and } 2 \le Y \le 4) \\ &= F(3,4) - F(2,4) - F(3,2) + F(2,2) \\ &= 1 - \frac{64}{156} - \frac{66}{156} + \frac{24}{156} = \frac{25}{78}. \end{aligned}$$

(c) Since y must lie in the interval $0 \le y \le 4$, $F_2(y) = 0$ for $y < 0$ and $F_2(y) = 1$ for $y > 4$. For $0 \le y \le 4$,

$$F_2(y) = \lim_{x\to\infty} F(x,y) = \lim_{x\to 3} \frac{1}{156} xy(x^2+y) = \frac{1}{52} y\,(9+y).$$

(d) We have $f(x,y) = 0$ unless $0 \le x \le 3$ and $0 \le y \le 4$. In this rectangle we have

$$f(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y} = \frac{1}{156}(3x^2 + 2y).$$

(e) The region over which to integrate is shaded in Fig. S.3.25.

$$Pr(Y \le X) = \int\int_{\substack{\text{shaded} \\ \text{region}}} f(x,y)\,dx\,dy = \int_0^3 \int_0^x \frac{1}{156}(3x^2+2y)\,dy\,dx = \frac{93}{208}.$$

9. The joint p.d.f. of water demand X and electricy demand Y is in (3.4.2), and is repeated here:

$$f(x,y) = \begin{cases} 1/29204 & \text{if } 4 \le x \le 200 \text{ and } 1 \le y \le 150, \\ 0 & \text{otherwise.} \end{cases}$$

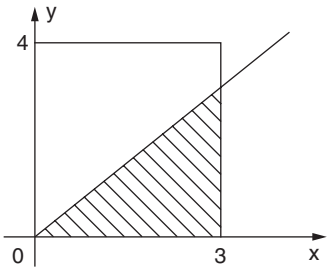


Figure S.3.25: Figure for Exercise 8e of Sec. 3.4.

We need to integrate this function over the set where $x > y$. That region can be written as $\{(x, y) : 4 < x < 200, 1 < y < \min\{x, 150\}\}$. The reason for the complicated upper limit on y is that we require both $y < x$ and $y < 150$.

$$\begin{aligned}
\int_4^{200} \int_1^{\min\{x,150\}} \frac{1}{29204} dy dx &= \int_4^{200} \frac{\min\{x-1, 149\}}{29204} dx \\
&= \int_4^{150} \frac{x-1}{29204} dx + \int_{150}^{200} \frac{149}{29204} dx \\
&= \left. \frac{(x-1)^2}{2 \times 29204} \right|_{x=4}^{150} + \frac{50 \times 149}{29204} \\
&= \frac{149^2 - 3^2}{58408} + \frac{7450}{29204} = 0.63505.
\end{aligned}$$

10. (a) The sum of $f(x, y)$ over all x for each fixed y is

$$\exp(-3y) \sum_{x=0}^{\infty} \frac{(2y)^2}{x!} = \exp(-3y) \exp(2y) = \exp(-y),$$

where the first equality follows from the power series expansion of $\exp(2y)$. The integral of the resulting sum is easily calculated to be 1.

(b) We can compute $\Pr(X = 0)$ by integrating $f(0, y)$ over all y:

$$\Pr(X = 0) = \int_0^{\infty} \frac{(2y)^0}{0!} \exp(-3y) dy = \frac{1}{3}.$$

11. Let $f(x, y)$ stand for the joint p.f. in Table 3.3 in the text for $x = 0, 1$ and $y = 1, 2, 3, 4$.

(a) We are asked for the probability for the set $\{Y \in \{2, 3\}\} \cap \{X = 1\}$, which is $f(1, 2) + f(1, 3) = 0.166 + 0.107 = 0.273$.

(b) This time, we want $\Pr(X = 0) = f(0, 1) + f(0, 2) + f(0, 3) + f(0, 4) = 0.513$.

3.5 Marginal Distributions

Commentary

Students can get confused when solving problems like Exercises 7 and 8 in this section. They notice that the functional form of $f(x, y)$ factors into $g_1(x)g_2(y)$ for those (x, y) pairs such that $f(x, y) > 0$, but they don't understand that the factorization needs to hold even for those (x, y) pairs such that $f(x, y) = 0$. When the